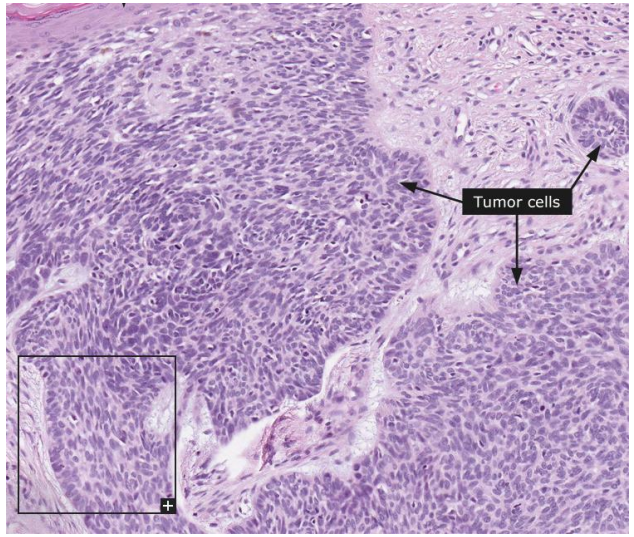
Motivation

Beyond mere prediction:

Asking additional questions

- Where?
- Why?
- How?
- …

Prediction: here are some cancerous cells,
additional question: why? What is different in those cells?



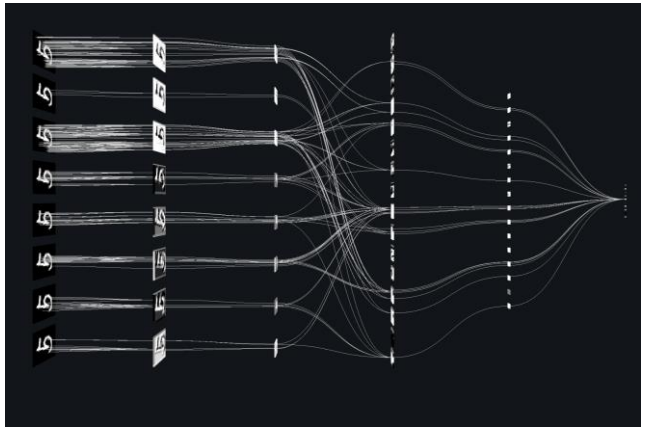
[source: Human Protein Atlas Project]

Prediction: cat,
additional question: where?



[source: Puzzles Brain via TipHero]

Prediction: digit 5,
additional question: how did you come up with that prediction?



Motivation

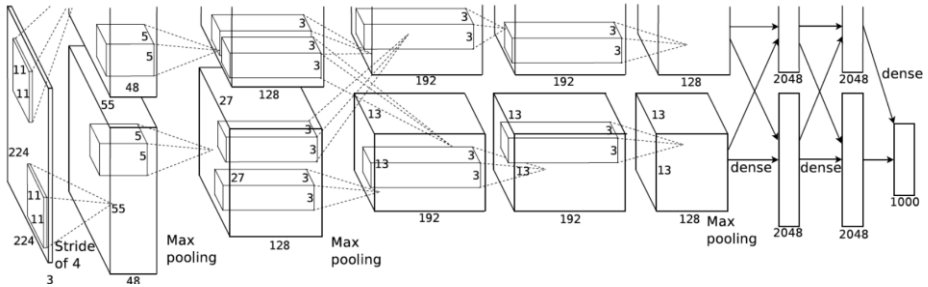
Ultimately, we would like the ML system to be able to *reason*

Motivation

Overparametrized Modeling

2012

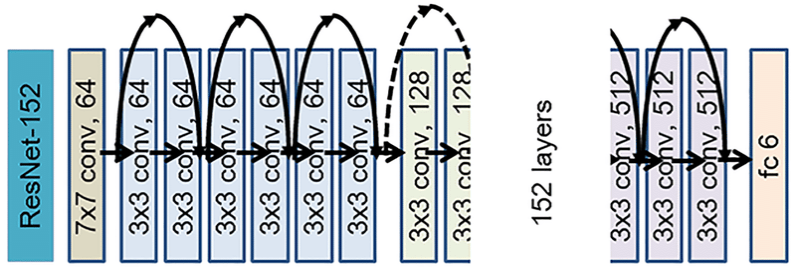
AlexNet



~60 million parameters

2020

ResNet 152x4



~1 billion parameters

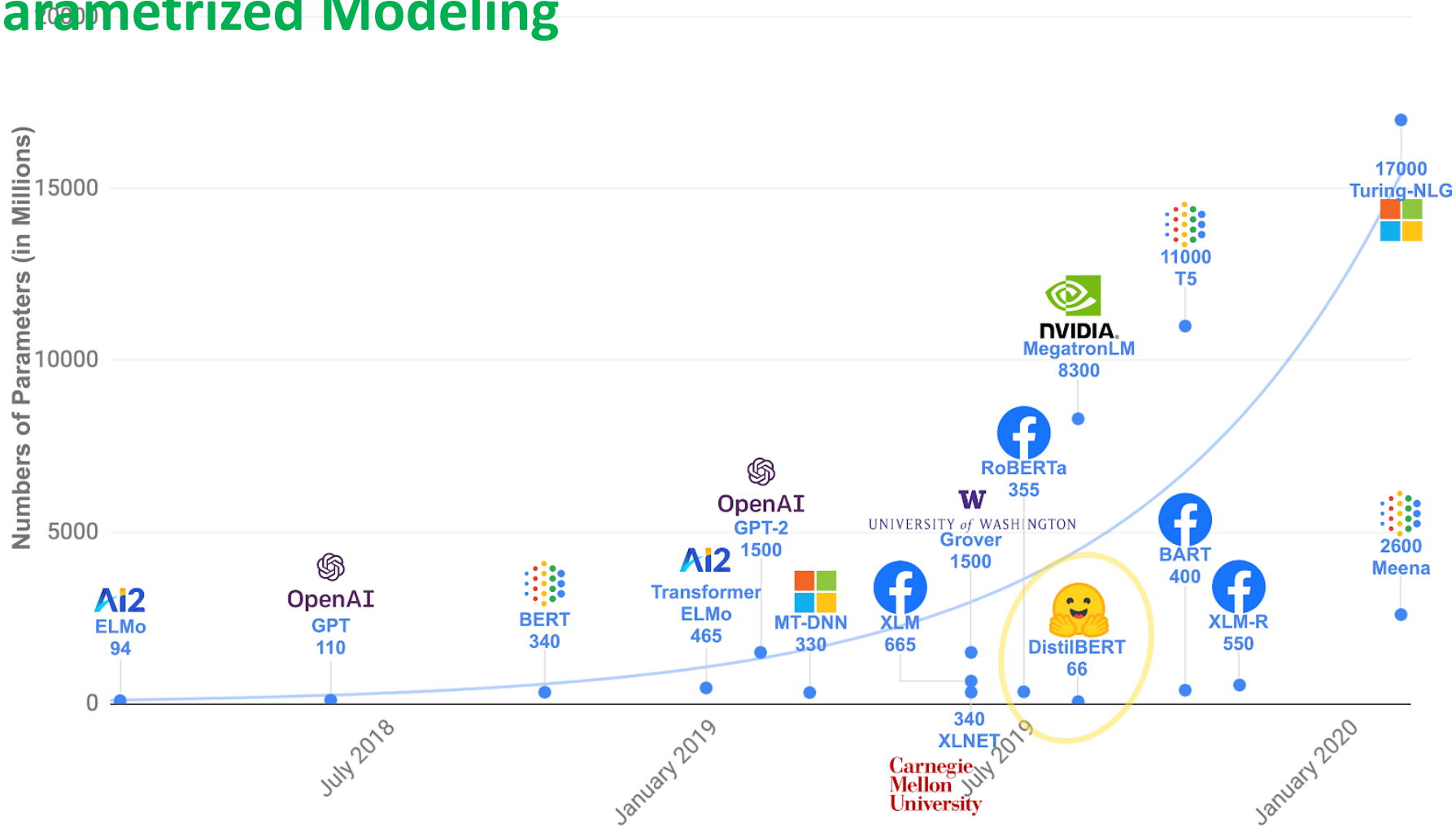
ImageNet ~1 million images

JFT dataset 300 million images

[Krizhevsky et al., "ImageNet Classification with Deep Convolutional Neural Networks", NIPS 2012]
[Kolesnikov et al., "Big Transfer (BiT): General Visual Representation Learning", ECCV 2020]

Motivation

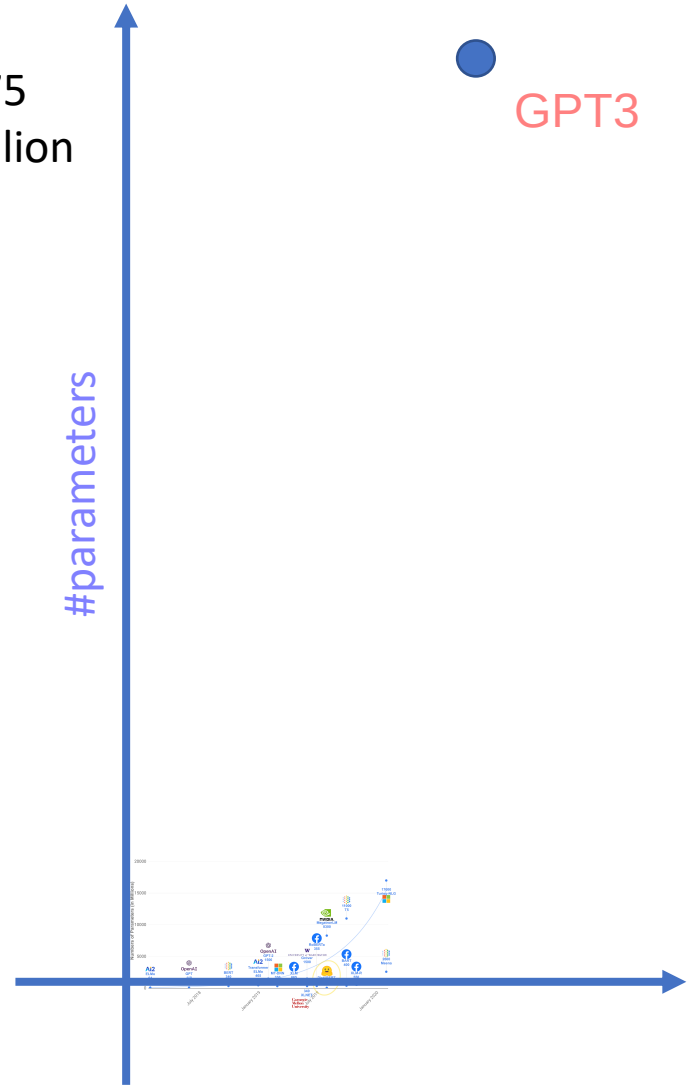
Overparametrized Modeling



[source: “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter”, 2019]

Motivation

Overparametrized Modeling

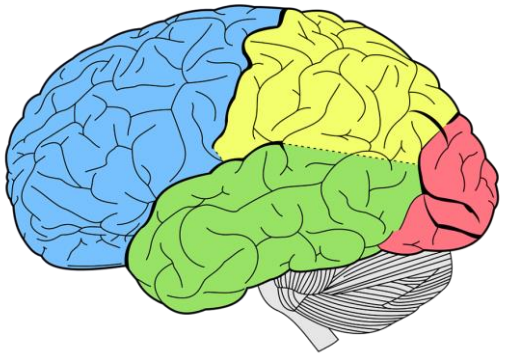


[Brown et al., "Language Models are Few-Shot Learners“, arXiv 2020]

Interpreting and Understanding Deep Networks

- largest networks today have as many as $10^{11} - 10^{12}$ parameters
- Our brain has $10^{14} - 10^{15}$ synapses





- NeuroScience
  - Studies the nervous system to understand the fundamental and emergent properties of neurons.
- Cognitive Psychology
  - Understanding the mental processes such as attention, language use, memory, perception, problem solving, creativity, and thinking.



